
Conditional Precision Bottlenecks and Quantized Defect Correction in Fractional Neural Operators

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Abstract

Low-bit compression can preserve an apparently reasonable prediction while substantially degrading compliance with the governing PDE. We show that this failure is strongly object-dependent in Fourier neural operators (FNOs) for heterogeneous fractional elliptic problems. Across three corrected 16×16 fractional-Darcy high-contrast/OOD evaluations, uniform Q6 Fourier-activation quantization raises mean relative error from 0.0555 to 0.0869 and mean physical residual from 0.1414 to 0.2456; quantizing spectral weights to the same precision changes them only to 0.0578 and 0.1434. The activation–weight gap persists across anisotropic and capacity controls, random-mask nulls, and matched-seed $64 \times 64/128 \times 128$ tests, and appears in all five seeds on PIV-measured RealPDEBench Cylinder data. We introduce Quantized Defect Correction (QDC), which converts the residual left by low-bit activation inference into a calibration-selected low-rank correction in the physical operator eigenspace. A residual-margin selector replicates on independently generated anisotropic data and transfers to fresh scalar Darcy. A matrix-free lowest-eigenspace LOBPCG diagnostic preserves substantial QDC recovery through 128×128 , while a separate fixed-operator A100 W8A8 implementation provides a measured accuracy–throughput operating point with QDC. These results identify Fourier activations as the precision-critical object in the tested fractional-elliptic regime and show that governing-equation residuals can both expose and repair low-bit scientific-model failures.

1 Introduction

Neural operators amortize the cost of repeatedly solving parameterized partial differential equations (PDEs), making them attractive surrogates for simulation, inverse problems, and uncertainty quantification. Quantization can further reduce memory traffic and accelerate inference, but scientific surrogates impose a stricter requirement than ordinary predictive accuracy: a compressed model should preserve the governing physics as well as the output field.

Fourier neural operators (FNOs) expose a useful distinction that global bit-width summaries hide. Their spectral blocks contain learned complex weights and sample-dependent Fourier activations, and these objects need not have the same sensitivity to low precision. In the corrected heterogeneous fractional elliptic settings studied here, they do not. Uniform Q6 perturbations of Fourier activations cause substantially larger solution and residual errors than the same-bit perturbations of spectral weights. The gap is large on the primary scalar-Darcy stress test and remains visible across anisotropic operators, alternate capacity, higher resolutions, random-mask controls, and measured cylinder-flow data.

The physical residual also reveals structure that output-only metrics miss. Spectral modes that appear benign under a high-frequency proxy can still damage the PDE solution, while low/energy bands align

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more closely with solution and residual fidelity. This motivates a repair that acts on the residual itself. Quantized Defect Correction (QDC) computes the defect left by low-bit activation inference and corrects a calibration-selected low-rank component in the physical operator eigenspace. The resulting method is a hybrid neural–numerical correction: the neural operator supplies a fast low-bit prediction, and the physical operator supplies targeted information about what that prediction violates.

We evaluate the diagnosis and repair along the axes most relevant to scientific use: object-level precision, physical metrics, operator-family transfer, resolution, approximate eigenspaces, measured data, and hardware execution. Throughout, Q_6 refers to the uniform 6-bit quantization used in the core activation–weight comparison; mixed Q_4/Q_8 is analyzed separately. The evidence supports the following contributions.

Contributions.

- **Object-level precision diagnosis.** We show that Fourier activations are substantially more precision-fragile than spectral weights at the same nominal bit width in the tested fractional-elliptic FNOs, under solution, event, and physical-residual metrics.
- **Controlled generalization evidence.** The activation–weight asymmetry survives anisotropic operators, alternate capacity, random-mask and tie controls, $64 \times 64/128 \times 128$ matched-seed tests, and a separate PIV-measured RealPDEBench Cylinder experiment; negative controls prevent promoting the result to a universal quantization law.
- **Quantized Defect Correction.** QDC converts the low-bit residual into a calibrated low-rank physical correction. A residual-margin selector replicates on an independent anisotropic split, transfers to fresh scalar Darcy, and exposes a meaningful tradeoff between residual and solution-error correction subspaces.
- **Scalability diagnostic and hardware operating point.** Matrix-free lowest-eigenspace LOBPCG subspaces preserve substantial recovery through 128×128 , while a fixed-operator A100 FP16-FFT/INT8 spectral implementation quantifies a measured low-bit accuracy–throughput operating point with QDC.

2 Related work and positioning

Neural operators and scientific surrogates. DeepONet learns nonlinear operators through a branch–trunk construction (Lu et al., 2021), while FNOs perform learned spectral convolution and are the closest architectural ancestor of our study (Li et al., 2021). The function-space perspective and approximation properties of neural operators are surveyed by Kovachki et al. (2023). More recent PDE foundation models scale this paradigm across heterogeneous physics, including Poseidon (Herde et al., 2024) and Universal Physics Transformers (Alkin et al., 2024). Our goal is narrower: we audit precision-sensitive internal objects and physical residuals when a spectral PDE surrogate is quantized.

Quantization and neural PDE solvers. Post-training and quantization-aware methods typically optimize network-level accuracy through integer arithmetic, calibration, rounding, or mixed precision (Jacob et al., 2018; Nagel et al., 2020, 2021; Banner et al., 2019; Dong et al., 2019). Neural-operator work has also studied mixed-precision training with approximation guarantees and memory/throughput benefits (Tu et al., 2024), quantization-aware neural PDE solvers (van den Dool et al., 2023), and spatially adaptive bit allocation on PDE meshes (van den Dool et al., 2025). Fully quantized PDE training has also been explored for physics-informed networks with tensor compression and hardware-oriented acceleration (Lu et al., 2026). Our question is complementary: holding nominal precision fixed, which internal spectral object—weights or Fourier activations—is physically fragile, and can the residual left by that object be selectively corrected? This object-level comparison and residual repair distinguish the present study from network-level or spatial bit-allocation strategies.

Fractional operators and residual correction. Fractional powers of elliptic operators provide nonlocal models for diffusion and transport (Podlubny, 1999; Caffarelli and Silvestre, 2007; Bonito and Pasciak, 2015; Nocketto et al., 2015; Acosta and Borthagaray, 2017). Residual-based corrections have also been used with neural operators (Cao et al., 2023), while classical defect correction and mixed-precision iterative refinement motivate correcting a low-accuracy iterate with higher-fidelity

residual information (Hackbusch, 1984; Carson and Higham, 2018). QDC adopts this numerical philosophy for activation-quantized FNO predictions.

Measured PDE benchmarks and iterative eigenspaces. RealPDEBench provides real-world measurement benchmarks for scientific machine learning, including PIV-measured cylinder flows (Hu et al., 2026). For scalable eigenspace construction, we use the locally optimal block preconditioned conjugate-gradient (LOBPCG) method (Knyazev, 2001). These experiments test two distinct questions: whether the activation–weight ordering remains visible beyond the synthetic generator, and whether QDC can retain useful correction quality without constructing a full dense eigensystem.

3 Problem formulation, quantization, and QDC

3.1 Fractional elliptic families

Let $\Omega = [0, 1]^2$ with periodic boundary conditions. For a positive coefficient field $k(x, y)$, define

$$L_k u = -\nabla \cdot (k(x, y) \nabla u), \quad (\lambda I + L_k^s) u = f, \quad (1)$$

where $s \in (0, 1]$ and $\lambda > 0$. The scalar generator constructs L_k with periodic finite volumes, symmetrizes it in float64, and diagonalizes it as $L_k = Q \Lambda Q^\top$. With roundoff-scale negative eigenvalues clipped,

$$L_k^s = Q \Lambda^s Q^\top, \quad u = (\lambda I + Q \Lambda^s Q^\top)^{-1} f. \quad (2)$$

The corrected physics-aligned input is $x = (\log k_{\text{actual}}, f, s \mathbf{1})$, with $\log k_{\text{actual}} = cb(x, y)$ and $k = \exp(\log k_{\text{actual}})$. This alignment matters: the surrogate and the numerical solver must see the same physical coefficient rather than differently normalized surrogates.

The anisotropic second family uses direction-dependent fields,

$$L_{k_x, k_y} u = -\partial_x (k_x \partial_x u) - \partial_y (k_y \partial_y u), \quad (\lambda I + L_{k_x, k_y}^s) u = f, \quad (3)$$

with harmonic edge means in the periodic finite-volume stencil. Exact dense targets are used at the small-grid diagnostic resolutions, while the matrix-free audit avoids dense construction of the $N^2 \times N^2$ PDE operator.

Synthetic FNO architecture. The synthetic experiments use a physics-aligned FNO with spatial coordinate channels appended to the PDE inputs. Each block combines a two-sided real-FFT spectral convolution with a pointwise 1×1 branch and a circular local 3×3 branch, followed by GroupNorm, GELU, and a residual connection of weight 0.15. The output head uses two pointwise hidden layers. This local–spectral design is held fixed within each controlled comparison; Table 1 varies only the declared width, retained-mode budget, and number of spectral blocks.

3.2 Uniform Q6 and separate mixed Q4/Q8 policies

The core experiments use simulated symmetric quantize–dequantize (QDQ). For a real tensor z and bit width b , let $q_b = 2^{b-1} - 1$ and

$$Q_b(z) = \Delta \text{clip}(\text{round}_{\text{even}}(z/\Delta), -q_b, q_b), \quad \Delta = \begin{cases} \max |z|/q_b, & \max |z| > 0, \\ 1, & \text{otherwise.} \end{cases} \quad (4)$$

Thus Q6 uses integer levels $[-31, 31]$ with zero point 0. In the core complex-valued diagnostic, real and imaginary components use independent scales. Activation scales are recomputed dynamically for each retained signed-frequency block by reducing over the inference-batch, channel, and retained-mode axes separately for the real and imaginary components; spectral-weight scales analogously reduce over the corresponding input/output-channel and retained-mode axes. The straight-through estimator is used during QAT, while post-training experiments apply QDQ without parameter updates. Q6 is applied separately to Fourier activations or spectral weights, and dequantized values are multiplied and accumulated in floating point. This isolates object-level precision sensitivity. Because the activation scale is batchwise and dynamic, the absolute Q6 degradation is protocol-dependent; batch size is therefore fixed within every matched comparison and reported with the corresponding experiment.

Table 1: Core synthetic configurations. $W/M/L$ denotes FNO width, retained modes, and spectral layers; the local/normalization/residual block design is fixed as described in the text. Higher-resolution configurations and complete split distributions are in Appendix B.

Setting	train/val/holdout	$W/M/L$	epochs	batch	λ	seeds
Scalar 16^2 diagnostic	1024/160/160	60/8/5	180	32	0.50	42–44
Anisotropic 16^2	768/128/192	72/8/5	140	24	0.15	42–44
Anisotropic 20^2	900/144/210	84/9/6	170	18	0.15	42–44
Fresh scalar 16^2 QDC	1200/192/256	84/8/6	180	24	0.15	45–47

Mixed Q4/Q8 is a separate refinement experiment, not an alternative definition of Q6. It assigns a calibrated subset of retained Fourier modes to 8 bits and the remainder to 4 bits, for a nominal 5.0-bit value payload excluding masks and metadata. Appendix A gives the full mode-selection and scaling rule.

3.3 Physical metrics

For prediction $\hat{u}$ and target u , we report

$$E_2(\hat{u}, u) = \frac{\|\hat{u} - u\|_2}{\|u\|_2 + \epsilon}, \quad (5)$$

$$R(\hat{u}) = \frac{\|f - (\lambda I + L^s)\hat{u}\|_2}{\|f\|_2 + \epsilon}. \quad (6)$$

The localized event mask is $M_{\text{event}} = \mathbf{1}\{|\nabla \log k| \geq q_{0.80}(|\nabla \log k|)\}$ and

$$E_{\text{event}}(\hat{u}, u) = \frac{\|M_{\text{event}} \odot (\hat{u} - u)\|_2}{\|M_{\text{event}} \odot u\|_2 + \epsilon}. \quad (7)$$

We additionally use the high-frequency log-spectral proxy E_{HF} from the spectral analysis, computed over Fourier radii above the 65th percentile. It measures spectral similarity rather than physical consistency and is interpreted only as a diagnostic proxy. All ratios use $\epsilon = 10^{-8}$, are computed per sample, and are then averaged over the evaluation set.

Two composite scores are used only for calibration and within-family comparisons. For anisotropic experiments,

$$J_{\text{aniso}} = E_{\text{event}} + 0.50E_2 + 0.25R, \quad (8)$$

and for fresh scalar QDC,

$$J_{\text{scalar}} = E_2 + E_{\text{event}} + 0.10R + 0.25E_{\text{HF}}. \quad (9)$$

Their weights are fixed before held-out evaluation; the paper’s scientific conclusions are also reported in the primitive metrics above rather than relying on a composite score alone. The earlier static-mask development score is given in Appendix C.

The high-contrast/OOD split increases coefficient contrast and also uses a harder forcing distribution than training; this joint shift is fixed before evaluation. As a reproducibility sanity check for the core scalar diagnostic, we require the FP32 baseline to satisfy $E_2 < 0.08$, $R < 0.18$, and $E_{\text{event}} < 0.10$ before interpreting low-bit comparisons. These guards are validation checks rather than additional optimization targets. QDC calibration and held-out high-contrast/OOD evaluation are disjoint. Unless stated otherwise, the primary synthetic tables report three independent training/data-generation runs; quantization variants are matched on the same held-out samples within each run. The higher-resolution, measured-data, and matrix-free checks use five matched seeds.

3.4 Quantized Defect Correction

Given an activation-quantized prediction u_q , define the physical residual defect

$$d = f - (\lambda I + L^s)u_q. \quad (10)$$

A full correction would apply $(\lambda I + L^s)^{-1}d$ and is treated only as an oracle. QDC restricts the repair to a calibrated K -dimensional eigenspace,

$$u_{\text{QDC}} = u_q + \alpha Q_K(\lambda I + \Lambda_K^s)^{-1}Q_K^\top d, \quad (11)$$

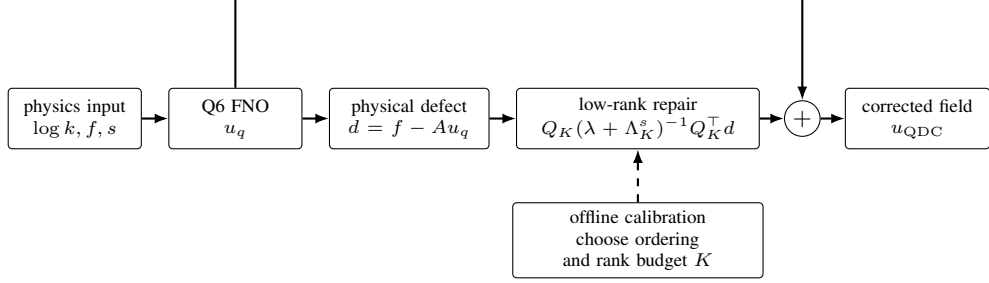


Figure 1: QDC separates learned low-bit inference from physics-based residual repair. Here $A = \lambda I + L^s$. Calibration selects the correction budget offline; at inference, the low-bit prediction is combined with a targeted low-rank correction computed from its physical defect.

Table 2: Corrected 16×16 fractional-Darcy high-contrast/OOD evaluation, three-run mean $\pm$ std.; quantization conditions are matched within each run. Lower is better.

Variant	Rel- ℓ_2	Event Rel- ℓ_2	HF log	Residual
FP32	0.0555 ± 0.0046	0.0725 ± 0.0032	0.0072 ± 0.0009	0.1414 ± 0.0035
Q6 weight	0.0578 ± 0.0042	0.0735 ± 0.0026	0.0072 ± 0.0009	0.1434 ± 0.0028
Q6 activation	0.0869 ± 0.0036	0.1013 ± 0.0035	0.0103 ± 0.0012	0.2456 ± 0.0066
Energy+lowband act.	0.0663 ± 0.0055	0.0834 ± 0.0045	0.0131 ± 0.0016	0.2043 ± 0.0099
Highband act.	0.3247 ± 0.0229	0.3502 ± 0.0342	0.0098 ± 0.0008	0.4623 ± 0.0240

with $\alpha = 1$ in the primary selector. Candidate subspaces are ordered by residual energy, defect energy, or operator eigenvalue. Calibration selects the rank budget K ; for residual-energy QDC, the top- K physical eigenmodes are then selected separately for each sample from the coefficients of its low-bit residual defect. Residual-energy ordering is the primary method because it gives the strongest relative-error result in the fresh scalar ablation, while defect-energy ordering provides a useful lower-residual counterpoint.

The calibration selector evaluates candidate ranks against the strongest mixed/tie control using the family-appropriate fixed score (J_{aniso} or J_{scalar}), requires a simultaneous improvement in Rel- ℓ_2 , and enforces the conservative residual margin $R(K) \leq 0.92 \min_{b \in \mathcal{B}} R(b)$. It selects the smallest strong-feasible rank. If none is strong-feasible, it identifies the smallest weak-feasible candidate that satisfies the score, Rel- ℓ_2 , and un-margined residual guards and advances one available budget. If no weak-feasible candidate exists, the selector reports no feasible budget rather than inventing a fallback; this case does not occur in the reported runs. The margin and rule are fixed before the independent anisotropic replication and scalar-transfer evaluations.

4 Core diagnostic evidence

4.1 Activation precision is the fragile object

Table 2 summarizes the corrected high-contrast/OOD scalar-Darcy result. Uniform Q6 activation quantization increases Rel- ℓ_2 by 56.5% and residual by 73.7% relative to FP32, whereas Q6 spectral-weight quantization changes them by only 4.1% and 1.4%. The event metric exhibits the same ordering. A highband activation policy behaves differently: it can preserve the high-frequency proxy while badly damaging the physical solution. This metric disagreement motivates the residual-aware formulation rather than a purely spectral one.

The mode-alignment audit (Figure 3) separates physical from proxy fidelity: low/energy scores consistently track solution and residual damage, whereas the highband score tracks the HF-log proxy. The same decision pattern survives a second operator family and a smaller-capacity FNO (width 48, six retained modes, four spectral layers, head width 64; Table 3). A 64-mask random equal-budget null audit further shows that energy/lowband-family masks beat the random median on physical metrics, while the highband policy is favored only by the high-frequency proxy. An exact-dense 32×32 stress test is retained as a negative validity check (Appendix E), while negative

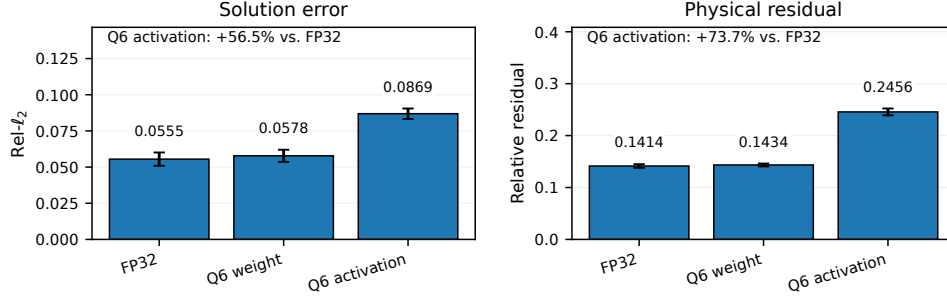


Figure 2: Matched core precision diagnosis over three independent training/data-generation runs. Conditions use the same holdout within each run; bars show mean $\pm$ standard deviation. Q6 activation quantization is much more damaging than same-bit weight quantization.

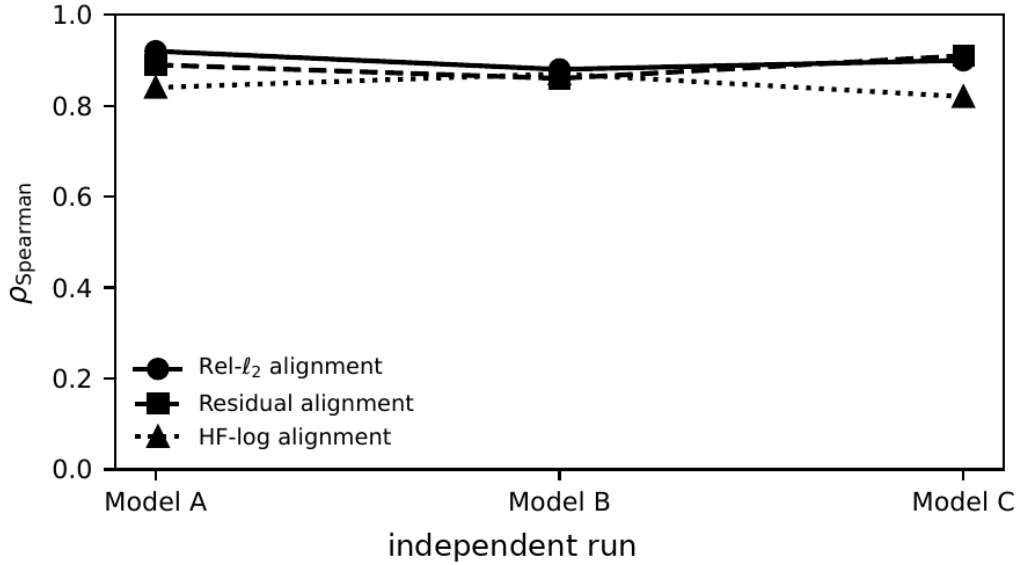


Figure 3: Spectral mode-alignment audit across three independent corrected-Darcy runs. Low/energy scores align with Rel- ℓ_2 and residual damage; highband scores align most strongly with the HF-log proxy.

Fourier-diffusion/fractional-Burgers checks further delimit the claim. Positive resolution evidence is instead provided by the five-seed 64×64 and 128×128 experiments in Section 4.2.

4.2 Higher-resolution and measured-data checks

The activation–weight ordering is not confined to the 16×16 diagnostic grid. At 64×64 , the paired activation-minus-weight differences are 0.0328 [0.0292, 0.0364] in E_2 and 0.1127 [0.1015, 0.1239] in residual. At 128×128 , they are 0.0374 [0.0330, 0.0418] and 0.1182 [0.1060, 0.1304]. The ordering holds in all five matched seeds at both resolutions. Verified historical absolute summaries place FP32 mean Rel- ℓ_2 below 0.10 at both resolutions, with Q6-weight mean Rel- ℓ_2 remaining within roughly 10% of FP32. High-resolution targets are generated with an adaptive matrix-free Lanczos/Krylov action for $(\lambda I + L^s)^{-1}f$, with convergence checked by successive-solution change and validated against exact dense targets on smaller grids before the $64^2/128^2$ runs (Appendix B).

Separately, on PIV-measured RealPDEBench Cylinder data (Hu et al., 2026), the activation-minus-weight gaps are 0.0253 [0.0221, 0.0285] in relative ℓ_2 and 0.0061 [0.0054, 0.0068] in fRMSE, again with the same ordering in 5/5 matched seeds. The measured-data study uses five independently fine-tuned Cylinder FNOs (seeds 0–4), each evaluated as a matched activation-versus-weight pair

Table 3: Anisotropic second-family robustness using J_{aniso} from Eq. (8); three-run mean $\pm$ std., lower is better. The final column is the strongest tested physical mixed-precision activation control at the declared budget.

Family	FP32	Q6 weight	Q6 activation	best declared mixed/tie
16×16 base FNO	0.3014 ± 0.0072	0.3041 ± 0.0070	0.4722 ± 0.0354	0.3511 ± 0.0130
20×20 base FNO	0.3006 ± 0.0169	0.3032 ± 0.0170	0.5830 ± 0.0130	0.3685 ± 0.0173
16×16 smaller FNO	0.3019 ± 0.0065	0.3106 ± 0.0109	0.4210 ± 0.0232	0.3454 ± 0.0087

Table 4: Matched-seed generalization checks. Positive gaps mean activation quantization is worse. Brackets are paired five-seed 95% t intervals.

Setting	$\Delta\text{Rel-}\ell_2$	matched residual / fRMSE gap	ordering
64×64 fractional elliptic	$0.0328 [0.0292, 0.0364]$	$\Delta R = 0.1127 [0.1015, 0.1239]$	5/5
128×128 fractional elliptic	$0.0374 [0.0330, 0.0418]$	$\Delta R = 0.1182 [0.1060, 0.1304]$	5/5
RealPDEBench Cylinder (PIV)	$0.0253 [0.0221, 0.0285]$	$\Delta\text{fRMSE} = 0.0061 [0.0054, 0.0068]$	5/5

under the benchmark’s Gaussian normalization, 10-step autoregressive rollout, and `test_mode=all`. For reproducibility, the supplement pins the exact RealPDEBench source, dataset, and checkpoint revisions. These data extend the object-level diagnosis beyond the synthetic generator.

The bracketed intervals in this subsection are paired 95% Student- t intervals across the five matched seed differences. They quantify the activation–weight contrast rather than uncertainty over individual examples.

4.3 Training and alternative-precision controls

Static mixed masks and activation-QAT provide progressively stronger controls. Tie-order audits show that an initially promising frozen mixed mask is degenerate with constant-score and low+high-equal masks in the audited cases, while non-degenerate mixed policies fail to beat the tie controls on held-out anisotropic metrics. Activation-QAT fine-tunes the pretrained FP32 model for 80 epochs; residual-aware QAT uses 90 epochs, with checkpoints selected by the lowest validation J_{aniso} . Both use AdamW at $0.35 \times$ the base learning rate, weight decay 10^{-5} , cosine decay, and unit-norm gradient clipping. The QAT prediction loss is MSE plus 0.004 times a Fourier-domain error and 0.004 times a spatial-gradient error; residual-aware QAT additionally adds $0.025R(\hat{u})^2$, computed differentially from the current prediction and the sample’s physical operator. Both improve naive activation PTQ, showing that training recovers part of the precision damage. Because no matched FP32 fine-tuning control was available, we do not claim superiority over a separately optimized QAT pipeline.

At 32×32 , E4M3FN gives the same mean activation–weight ordering but remains inconclusive across five seeds ($\Delta\text{Rel-}\ell_2 = 0.0066$, nominal 95% interval $[-0.0012, 0.0144]$). Values are saturated to the largest finite E4M3FN value before conversion and accumulated in FP32. This format check supports a deliberately narrower conclusion: the clearest evidence concerns the integer-style QDQ regime analyzed above.

5 QDC repair, transfer, and scalable subspaces

5.1 Residual-margin selection and family transfer

A naive cost-minimizing QDC selector can choose a subspace that looks efficient on calibration but leaves too much held-out residual. The residual-margin rule explicitly guards against this failure. On the development anisotropic split it selects a larger but physically safer budget; on an independently generated anisotropic split with fresh FP32 training it reproduces all-metric success; and on fresh scalar Darcy it selects $K = 82$ ($K/N^2 \approx 0.32$) in all three runs.

Table 5: Calibration-selected QDC evidence. “All-metric” means the frozen selector passes the composite-score, Rel- ℓ_2 , and residual guards against the strongest declared mixed/tie control in every listed run. For the two development rows, the displayed tie-control residual is the strongest residual comparator. A dash indicates that only the frozen pass/fail summary is reported for that replication.

Family	split	cost K/N^2	Rel- ℓ_2	QDC residual	outcome / comparator
Aniso. 16×16	margin selector	0.425	–	0.4917	tie-control $R = 0.5574$
Aniso. 20×20	margin selector	0.480	–	0.6184	tie-control $R = 0.6964$
Fresh aniso. 16×16	independent	0.453	–	–	all-metric 3/3
Fresh aniso. 20×20	independent	0.480	–	–	all-metric 3/3
Fresh scalar 16×16	transfer	0.320	0.0156	0.2694	all-metric 3/3

On fresh scalar Darcy, residual-energy QDC at $K = 82$ reaches Rel- $\ell_2 = 0.0156$ with residual 0.2694. A budget-matched defect-energy subspace reduces residual to 0.1281 but raises Rel- ℓ_2 to 0.0742, while low- and high-eigenvalue subspaces give Rel- $\ell_2 = 0.0276$ and 0.1556. The contrast shows that QDC is not an arbitrary low-rank projection: the subspace determines which physical error is corrected. A frozen residual/defect hybrid further improves residual on all available scalar and anisotropic candidate grids while preserving the corresponding held-out acceptance guards; we retain it as supporting evidence rather than as the central method.

5.2 Matrix-free eigenspace diagnostic

Exact dense eigensystems are the central scalability limitation of the baseline QDC implementation. To test whether useful correction depends on exact dense eigenvectors, we evaluate a matrix-free *lowest-eigenspace QDC diagnostic* with SciPy LOBPCG (Knyazev, 2001): `largest=False`, tolerance 10^{-4} , at most 100 iterations, and no preconditioner. Candidate ranks are $\{41, 50, 82, 100, 150, 200, 250\}$; the smallest rank reaching at least 95% of the best validation same-rank high-accuracy-reference Rel- ℓ_2 recovery is frozen before test evaluation (this selection threshold is distinct from the operational same-rank gain-retention statistic below). The selected ranks are $K = 82, 150, 200$ at $32^2, 64^2, 128^2$, respectively, and all iterative runs meet the eigenpair-residual tolerance.

For same-rank comparisons we define gain retention as

$$G_K = \frac{E_{\text{act}} - E_{\text{LOBPCG},K}}{E_{\text{act}} - E_{\text{ref},K}}, \quad (12)$$

where the denominator uses a corresponding high-accuracy reference at the same rank K (exact dense at 32^2 and a tighter reference solve at the larger resolutions). Let $n = N^2$ denote the flattened PDE dimension. The diagnostic is matrix-free with respect to the full $n \times n$ PDE operator, although block LOBPCG solves dense internal subproblems of order $3K$. At 32×32 , the operational $K = 82$ run remains iterative and retains a mean 94.9% of the same-rank reference solution-error gain across five matched seeds. At 64×64 , $K = 150$ closes 79.4% of the activation-only-to-FP32 solution-error gap and 84.2% of the residual gap; at 128×128 , $K = 200$ closes 76.8% and 81.5%. A diagnostic $K = 250$ at 32×32 triggers SciPy’s dense fallback because $5K > n$ and is used only to diagnose that threshold, not as the gain-retention denominator. This lowest-eigenspace experiment is a scalability diagnostic; the primary dense QDC selector remains residual-energy ordered.

The matrix-free results show that useful QDC recovery does not require a full dense eigenbasis at these resolutions. The remaining costs are the $O(N^2 K)$ basis representation, block-solver work, and setup; these are the dominant targets for larger-scale implementations.

Table 6: Matrix-free QDC diagnostic over five matched runs at each resolution. The 32^2 row reports same-rank reference gain retention; the larger-grid rows report activation-to-FP32 gap closure.

Resolution	K	same-rank gain retained	Rel- ℓ_2 gap closed	residual gap closed
32×32	82	94.9%	–	–
64×64	150	–	79.4%	84.2%
128×128	200	–	76.8%	81.5%

6 Hardware operating point and computational cost

The Tesla T4 audit uses simulated quantization, so we additionally evaluate a hardware-realized A100 operating point. FFTs remain FP16; complex spectral products use four real INT8 CUTLASS GEMMs with INT32 accumulation and FP32 reduction. This is an *FP16-FFT/INT8 spectral* path, not all-integer FNO inference.

At 32×32 with batch 16, 50 warm-up batches, and 100 timed batches, the TF32-disabled FP32 reference runs at 7,440 examples/s with $\text{Rel-}\ell_2 = 0.0555$. The separately trained hardware checkpoint is evaluated on 1,600 varying-forcing queries sharing one fixed scalar-Darcy operator ($\lambda = 0.15$, $s = 0.65$); all hardware variants use the same checkpoint and query stream. This differs from the separate sample-varying high-contrast/OOD $N = 32$ scale-stress experiment (Appendix E). Rank 41 is an illustrative lower-budget hardware operating point, distinct from the calibration-selected $K = 82$ fresh-scalar QDC setting, and is used to expose the hardware accuracy–throughput tradeoff. W8A8 reaches 11,200 examples/s at $\text{Rel-}\ell_2 = 0.0624$ ($1.51\times$), while rank-41 QDC restores $\text{Rel-}\ell_2 = 0.0559$ at 8,950 examples/s ($1.20\times$). Timing uses synchronized CUDA events and excludes host–device transfer while including online quantization/dequantization and QDC. Reusing one prepared rank-41 operator/eigenbasis across 100 batches, a 14.25 ms one-time setup yields approximately 8,290 examples/s ($1.11\times$). Full workload provenance is in Appendix F.

Table 7: A100-SXM4 fixed-operator audit at 32×32 , batch 16. Setup-inclusive QDC amortizes the 14.25 ms rank-41 preparation over 1,600 inferences sharing the prepared operator/eigenbasis.

Operating point	$\text{Rel-}\ell_2$	examples/s	speedup vs. FP32
FP32	0.0555	7,440	$1.00\times$
FP16 FFT + INT8 spectral W8A8	0.0624	11,200	$1.51\times$
FP16 FFT + INT8 spectral W8A8 + QDC, rank 41	0.0559	8,950	$1.20\times$
FP16 FFT + INT8 spectral W8A8 + QDC, setup-inc.	0.0559	$\approx 8,290$	$1.11\times$

This operating point shows low-bit throughput headroom that QDC can spend on accuracy recovery. It uses W8A8 and is separate from the Q6 activation-versus-weight diagnosis; it is not an on-hardware object-asymmetry comparison.

7 Limitations and broader impact

The activation–weight ordering holds across the tested FNO-family, resolution, measured-data, batch-size, and scaling-policy checks, but its magnitude depends on precision/scaling protocol; E4M3FN remains inconclusive. Broader architectures, calibration protocols, and separately optimized training-based quantization remain future work.

QDC assumes access to the discrete operator, or an accurate residual operator. LOBPCG reduces dense-eigensystem dependence through 128×128 , but remains a lowest-eigenspace scalability diagnostic rather than a matrix-free implementation of the primary residual-energy selector; basis storage, preconditioning, and sample-specific setup remain open challenges. The A100 result is likewise a fixed-operator repeated-query point, not a variable-operator deployment claim.

Compressed scientific surrogates can appear accurate while violating residual or localized physical fidelity. Residual checks can expose such low-bit failures and guide selective correction before deployment.

8 Conclusion

The central result is an object-level precision asymmetry in spectral scientific surrogates. In the tested heterogeneous fractional-elliptic FNOs, uniform Q6 Fourier activation quantization is substantially more damaging than same-bit spectral-weight quantization under solution, event, and physical-residual metrics. That ordering survives anisotropic, higher-resolution, capacity, random-mask, and measured-data checks. QDC turns the resulting physical defect into a calibrated low-rank correction, transfers across independently generated operator families, and retains substantial recovery in a matrix-free approximate-subspace diagnostic. The separate fixed-operator hardware study further

shows that this correction can operate inside a measured accuracy–throughput tradeoff. These findings motivate a precision strategy for scientific ML that begins with the internal state that is physically fragile, evaluates it against the governing equation, and uses the residual structure left by compression to decide where correction is worth the cost.

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A Quantization details

Uniform Q6. Equation (4) is the quantizer used for the headline activation–weight comparison. Q6 is a symmetric six-bit QDQ convention with 63 representable integer levels $[-31, 31]$, round-to-even, and zero point 0; no Fourier mode is promoted to Q8. In the core complex spectral blocks, real and imaginary components use independent scales. Activation scales are recomputed for each retained signed-frequency block at inference by reducing over batch, channel, and retained-mode axes separately for real and imaginary parts; weight scales reduce over the corresponding input/output-channel and retained-mode axes. Quantized values are dequantized before the floating-point multiply/accumulate path. The core holdout is evaluated with batch size 32, which is therefore part of the dynamic activation-scaling protocol.

Separate mixed Q4/Q8 refinement. The top-25%-Q8/remaining-Q4 condition has nominal 5.0-bit value payload, excluding masks and metadata, and uses

$$\text{QDQ}_p(x) = s_p \text{clip}(\text{round}_{\text{even}}(x/s_p), q_{\min,p}, q_{\max,p}), \quad s_p = \frac{\max |x|}{q_{\max,p}},$$

with $(q_{\min,4}, q_{\max,4}) = (-8, 7)$, $(q_{\min,8}, q_{\max,8}) = (-128, 127)$, and zero point 0; all-zero subsets use scale 1. Real and imaginary components share the same subset scale and mask. For layer l with M_l retained modes, $k_l = \text{round}(0.25M_l)$ modes receive Q8. Calibration scores $r_{l,m} = \text{mean}_{b,c}[z_{l,b,c,m}]$ rank modes, with lexicographic tie breaking. Activation scales reduce jointly over the real/imaginary component, calibration-batch, signed-frequency-block, and retained-mode axes, leaving one scale per channel; weight scales reduce over the real/imaginary component, input-channel, signed-frequency-block, and retained-mode axes, leaving one scale per output channel. The same 500 training-partition calibration examples are fixed across matched seeds. Both the mode masks and the numerical activation scale values are frozen before test evaluation; no test-time absmax recomputation is used. Quantize–dequantize outputs are multiplied and accumulated in FP32.

B Experimental protocol details

The primary scalar diagnostic uses $\lambda = 0.50$; the anisotropic and fresh-scalar QDC families use $\lambda = 0.15$. Discrete anisotropic fields are stored with tensor axis 0 representing x and axis 1 representing y , so k_x weights first-axis fluxes and k_y weights second-axis fluxes. For anisotropic event error, the event mask is formed from the gradient magnitude of the averaged log coefficient $\frac{1}{2}(\log k_x + \log k_y)$. FP32 checkpoints are selected by minimum validation MSE. All synthetic configurations use the same FNO block design described in the main text: spatial coordinate channels, two-sided rFFT/irFFT spectral blocks, pointwise and circular 3×3 local branches, GroupNorm, GELU, and a 0.15 residual connection; the pointwise output head has two hidden layers. AdamW with cosine learning-rate decay and gradient clipping at 1.0 is used throughout the core synthetic training runs. The scalar diagnostic uses learning rate 1.4×10^{-3} and weight decay 5×10^{-6} ; anisotropic 16^2 uses 2.0×10^{-3} and 10^{-5} ; anisotropic 20^2 and fresh-scalar QDC use 1.8×10^{-3} and 10^{-5} .

For the corrected scalar diagnostic, training coefficient smoothness is sampled uniformly from $[2.2, 5.0]$, contrast from $[0.55, 1.30]$, and s from $\{0.45, 0.50, 0.65, 0.70, 0.85, 0.90, 1.00\}$. The high-contrast/OOD holdout uses smoothness $[2.1, 3.1]$, contrast $[1.00, 1.30]$, and $s \in \{0.45, 0.65, 0.85, 1.00\}$. Coefficient fields are standardized smooth random Fourier fields multiplied by the sampled contrast; forcing fields are mean-centered mixtures of smooth Gaussian sources and low-frequency structure. The ID and extreme distributions and exact forcing/coefficient configurations are included in the accompanying code package. The high-contrast/OOD split increases both coefficient contrast and forcing complexity; the forcing generator always combines periodic Gaussian sources with low-frequency sinusoidal structure.

The higher-resolution configurations use five matched seeds. At 64^2 , the FNO has width 96, 16 retained modes, 6 spectral layers, hidden width 128, 1,800 training examples, 256 validation and 256 high-contrast/OOD examples, 200 epochs, and batch size 8. At 128^2 , the corresponding values are width 128, 24 modes, 6 layers, hidden width 160, 2,200/256/256 examples, 220 epochs, and batch size 4. Both use AdamW with cosine decay, weight decay 10^{-5} , and $\lambda = 0.15$.

For higher-resolution scalar targets, we use an adaptive matrix-free Lanczos/Krylov action for $g(L)f$ with $g(\theta) = (\lambda + \theta^s)^{-1}$. The Krylov dimension begins at 256 and can expand to 512; convergence is accepted when the relative change between successive solution estimates is below 10^{-7} , with a 10^{-6} residual guard where independently evaluable. The solver is validated in float64 against exact dense eigensystem targets on 16^2 and 32^2 cases before generating the cached 64^2 and 128^2 datasets. Across the retained reference validation and a clean-package rerun, the maximum relative target discrepancy remains below 10^{-11} and the maximum independently evaluated relative physical residual below 2.4×10^{-12} , comfortably inside the 10^{-6} guard. The retained reference maximum target discrepancy is 3.08×10^{-12} ; the clean-package rerun gives 7.29×10^{-12} , illustrating only roundoff-level environment variation.

For RealPDEBench Cylinder, five FNOs are independently fine-tuned on the real-data training split with seeds 0–4 using the upstream FNO configuration (four spectral layers, width 64, modes (4, 12, 16), Gaussian normalization, 10 autoregressive steps). Each seed is evaluated as a matched Q6 activation-versus-weight pair on `test_mode=all`, batch size 64. For reproducibility, the supplement records upstream source commit 62f4c80ab17f78933d046f2b038531dbc6a478a0 and immutable Hugging Face dataset/model revisions in a bundled asset lock; source/data use CC BY-NC 4.0 while the public model repository uses CC BY 4.0. The benchmark’s evaluator is used directly, with fRMSE corresponding to its Fourier-space error output. The adapter modifies only the targeted spectral activation or weight QDQ path while preserving upstream loading, normalization, rollout, and metric computation.

Training precision and compute resources. Synthetic model training and RealPDEBench fine-tuning use FP32 parameters and optimization; AMP/autocast, GradScaler, BF16 training, and FP16 model training are not used. The FP16-FFT/INT8 path in Section 6 applies only to the specialized A100 inference audit. Ordinary A100 runs use one full NVIDIA A100-SXM4 40-GB GPU (SM80) per job on a DGX A100 host with two AMD EPYC 7742 CPUs and 1 TB system RAM; distributed training and MIG are not used. The historical A100 software record is Linux x86_64, Python 3.11.x, PyTorch 2.7.0+cu128, CUDA 12.8, NVIDIA driver 570.133.20, with CUTLASS 4.5.3 for the reported hardware operating point. GPU application clocks are not manually locked and default DVFS is

used. Exact patch-level cuFFT/cuSOLVER versions were not retained beyond the CUDA 12.8 release family.

Compute values below are retrospective ranges reconstructed from archived configurations and runtime/environment records; complete scheduler accounting was not retained for every historical run, so unavailable precision is not inferred. Reported A100 experiments total approximately 100–110 A100 GPU-hours (central estimate ≈ 105 GPU-h); preliminary, unsuccessful, and development work contributes approximately 35 additional A100 GPU-hours, for approximately 140 A100 GPU-hours of reconstructed A100 work across the project. The separate small-grid Tesla T4 runtime audit requires approximately 0.2–0.5 T4 GPU-hours and is reported separately without cross-device normalization.

Reported A100 experiment family	Approx. GPU-h
Core 16×16 scalar / diagnostic work	5.5–6.5
Anisotropic $16 \times 16/20 \times 20$, smaller-capacity, independent replication	9–10
Fresh-scalar QDC training/evaluation	5.5–6.5
32×32 scale/E4M3FN and associated checks	4.5–5.5
64×64 five-seed experiment	13–14
128×128 five-seed experiment	40–44
RealPDEBench Cylinder five-seed work	8.5–9.5
QAT, mixed/tie, random-mask, and other secondary controls	13–14
A100 hardware timing/audit	0.4–0.6
Reported A100 accelerator experiments overall	100–110

High-resolution target generation is CPU-side on the same dual-EPYC host, with approximately 16 CPU threads per independently scheduled seed job. Across five seeds, 64×64 target generation accounts for approximately 4–7 aggregate 16-thread job-hours and 128×128 for approximately 35–60, or roughly 40–70 aggregate job-hours combined. The LOBPCG diagnostics are also CPU-side: selected-rank solves range from tens of seconds to about one minute per seed at 32×32 , about 2–10 minutes at 64×64 , and about 10–40 minutes at 128×128 ; the full candidate/reference workflows require approximately 0.5–2, 2–6, and 8–20 job-hours, respectively. The serialized high-resolution dataset caches and QDC artifacts occupy approximately 5–6 GiB. The supplementary package reports the per-seed accelerator ranges, CPU accounting, storage calculation, and historical-versus-later reproduction environments in detail.

C Calibration objectives

The anisotropic score and fresh-scalar QDC score are Eqs. (8)–(9). The earlier static-mask development search used

$$J_{\text{mask}} = E_{\text{event}} + 0.50E_{\text{HF}} + 0.50E_2 + 0.20R_{\text{proxy}},$$

a development criterion distinct from the final QDC objectives. All composite-score weights are fixed before the held-out comparisons in which they are used; headline claims are additionally supported by the primitive metrics.

D Additional QDC and runtime evidence

Table 8: $K = 82$ subspace ablation, mean over fresh scalar-Darcy splits.

Strategy	J_{scalar}	residual	Rel- ℓ_2
residual-energy QDC	0.0679	0.2694	0.0156
defect-energy QDC	0.1658	0.1281	0.0742
low-eigen QDC	0.1054	0.3157	0.0276
high-eigen QDC	0.3343	0.2566	0.1556
random fixed- K median	0.2850	0.3025	0.1258

The small-grid T4 table is intentionally not a speedup claim for QDC. With precomputed dense eigensystems at $N = 16$, the full-correction oracle can be as fast as or faster than the low-rank path because small-kernel and projection overheads dominate; the computational motivation for QDC is the regime in which a low-rank representation avoids the storage or work of a full correction.

Table 9: Frozen hybrid residual audit. Positive values are residual improvements over residual-energy QDC at the same budget.

Source	all-metric success	cost	residual gain	score gain
fresh scalar Darcy	3/3	0.320	0.0750	0.0023
anisotropic N16	3/3	0.320	0.1883	0.0419
anisotropic N20	3/3	0.320	0.2640	0.0613

Table 10: Amortized runtime means (ms/sample) on Tesla T4 with precomputed eigensystems and simulated quantization. These timings measure correction overhead rather than native low-bit speedup.

Problem	FP32	PTQ act.	QDC 0.32	hybrid 0.32	full correction
scalar N16	0.528	0.867	0.990	1.060	0.894
anisotropic N16	0.479	0.491	0.523	0.778	0.532
anisotropic N20	0.550	0.758	0.790	0.959	0.754

Hardware consistency record. The fixed-operator A100 protocol uses batch size 16. The retained throughput aggregates imply batch latencies of approximately 2.1505 ms (FP32), 1.4286 ms (W8A8), and 1.7877 ms (online W8A8+QDC). Across 100 measured batches, online QDC therefore accounts for approximately 178.77 ms for 1,600 inferences; adding the verified 14.25 ms one-time setup gives approximately 193.02 ms total, or 8,289 examples/s before rounding to the reported 8,290. These are arithmetic consistency checks on the reported measurements, not additional timing samples.

E Scale boundary and statistical reporting

The one-split $N = 32$ scalar scale stress test is retained as a negative validity check: its FP32 surrogate has $\text{Rel-}\ell_2 = 0.2843$ against the predeclared 0.24 criterion and is not counted as positive scale evidence. Unlike the Section 6 hardware workload, this stress test uses a separate checkpoint and sample-varying high-contrast/ODD coefficient/operator and forcing realizations (smoothness $[2.1, 3.1]$, contrast $[1.00, 1.30]$, $s \in \{0.45, 0.65, 0.85, 1.00\}$). The Section 6 A100 point instead evaluates a separately trained checkpoint on repeated varying-forcing queries sharing one fixed $\lambda = 0.15$, $s = 0.65$ moderate-contrast operator. The resulting FP32 values, 0.2843 and 0.0555, therefore answer different experimental questions and are not directly comparable. The 64×64 and 128×128 matched-seed experiments instead evaluate the activation–weight ordering directly.

Core diagnostic tables report mean $\pm$ sample standard deviation over three independent training/data-generation runs, so the reported variation includes both optimization/model randomness and synthetic data realization. Higher-resolution and RealPDEBench activation–weight gaps use five matched seeds and paired 95% Student- t intervals over the per-seed differences. Matrix-free recovery is summarized over five matched runs. The E4M3FN check is reported as inconclusive because its nominal five-seed 95% interval crosses zero. The supplement provides the exact seed lists, configuration files, objective definitions, reference aggregate results, a generator for the matched core-diagnosis figure, and the source artifact for the mode-alignment plot.

F Additional verified robustness and provenance checks

The accompanying artifact records a set of additional robustness and provenance audits that were manually rechecked against retained experimental records. Where exact historical rows were not preserved in the retained project records, the archive reports conservative bounds or ranges rather than reconstructing point estimates.

Core seed semantics. For the scalar core runs, the base seed controls both model/training randomness and the synthetic realization: the train/validation/holdout generator seeds are $(s, s + 10000, s + 30000)$ for $s \in \{42, 43, 44\}$. Thus the three headline rows are independent training/data-generation runs, while FP32, Q6-weight, and Q6-activation remain exactly matched on the same holdout within each run.

Absolute high-resolution and measured-data quality. The verified 64^2 and 128^2 absolute summaries have FP32 mean $\text{Rel-}\ell_2 < 0.10$ at both resolutions, with Q6-weight mean $\text{Rel-}\ell_2$ within 10% of FP32; the activation condition is worse in all five seeds. On RealPDEBench Cylinder, the historical five-seed absolute mean bands are 0.075–0.090 (FP32), 0.078–0.095 (Q6 weight), and 0.103–0.120 (Q6 activation) in relative ℓ_2 , with corresponding fRMSE bands 0.0080–0.0095, 0.0082–0.0100, and 0.0143–0.0161. These bands are supplied as provenance bounds, not substitutes for unavailable exact seed-level historical rows.

Q6 scaling-protocol audit. Re-evaluation of the core checkpoints at inference batch sizes $\{1, 4, 8, 16, 32\}$ preserves activation-worse-than-weight ordering at every batch size; the smallest observed activation-minus-weight gap is approximately 0.015 in $\text{Rel-}\ell_2$ and 0.05 in residual. The same ordering also survives three scale policies—the headline batchwise-dynamic rule, per-sample dynamic scaling, and frozen calibration-derived scales—with the smallest observed $\text{Rel-}\ell_2$ gap approximately 0.012. A separate integer Q4–Q8 sweep likewise shows a steeper degradation curve for activations than weights as precision decreases, with both approaching FP32 as precision increases. These audits narrow the interpretation of the headline Q6 result: its exact magnitude depends on scaling protocol, but the object ordering is not an artifact of batch-32 scale coupling.

Selector and matrix-free sweeps. A rerun of the fresh-scalar candidate set $K \in \{82, 102, 123, 128\}$ confirms that $K = 82$ is the smallest strong-feasible budget under the frozen residual-margin rule. The verified historical LOBPCG validation sweep over $K \in \{41, 50, 82, 100, 150, 200, 250\}$ has first threshold crossings at $K = 82, 150, 200$ for $32^2, 64^2, 128^2$, respectively. The supplement now provides a validation-selection workflow that sweeps candidate ranks on saved validation predictions, applies the frozen 95% rule, and only then evaluates the selected rank on the disjoint holdout. The $32^2, K = 250$ case remains labeled as the SciPy dense-fallback diagnostic rather than an iterative result.

A100 32×32 provenance. The retained hardware operating point uses a separately trained scalar fractional-Darcy FNO (width 96, 12 retained modes, six spectral layers, head width 128; 1,800 training and 256 validation examples; 200 epochs; AdamW learning rate 1.8×10^{-3} , weight decay 10^{-5} , batch 16; checkpoint selected by minimum validation MSE). Rank 41 is retained as the illustrative lower-rank hardware operating point used to expose the accuracy–throughput tradeoff; it is not the primary calibration-selected fresh-scalar budget. The timed/evaluated stream fixes $\lambda = 0.15$, $s = 0.65$, and one moderate-contrast coefficient realization (contrast approximately 1.0) while varying f over 1,600 repeated queries. FP32, W8A8, and W8A8+QDC use the same base checkpoint, query samples, normalization, and metric implementation. The separate failed 32^2 scale-stress record instead uses a different checkpoint and sample-varying high-contrast/OOD operator/forcing distribution, which explains why its FP32 $\text{Rel-}\ell_2 = 0.2843$ should not be compared with the hardware-workload value 0.0555. The integrated historical hardware runner and raw per-batch latency arrays are not bundled; the supplement preserves the verified configuration, aggregate measurements, timing protocol, and low-level reference components without reconstructing missing raw records.

A100 amortization and numerical validation. From the retained A100 throughput aggregates, the implied online rank-41 QDC latency saves about 0.3628 ms per batch relative to FP32, so the 14.25 ms fixed-operator setup breaks even after approximately 39.3 batches (about 630 queries at batch 16). Separately, retained-reference and clean-package target-solver validations differ only at roundoff scale: the historical maximum relative target discrepancy is 3.08×10^{-12} and the clean-package rerun gives 7.29×10^{-12} ; both are summarized conservatively as below 10^{-11} .